

LPWAN-based emissions monitoring for public-sector environmental compliance and air quality enforcement.

Public / Government sector × Emissions and environmental monitoring × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS133 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
55 is the world moving	51 can we play, can we win	MEDIUM 0 named in the evidence	€220m observed evidence · per year	NOW budgeted procurement within 1...	L1 5 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is a managed LPWAN-based emissions monitoring service for public-sector environmental agencies and municipalities, enabling continuous air quality and emissions compliance tracking. It exists now because regulators worldwide are tightening mandatory emissions reporting and enforcement, as seen in Thailand, Nigeria, and European tenders.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€220m €220m – €482m	€220m €220m – €482m	€2.6m €2.6m – €5.8m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

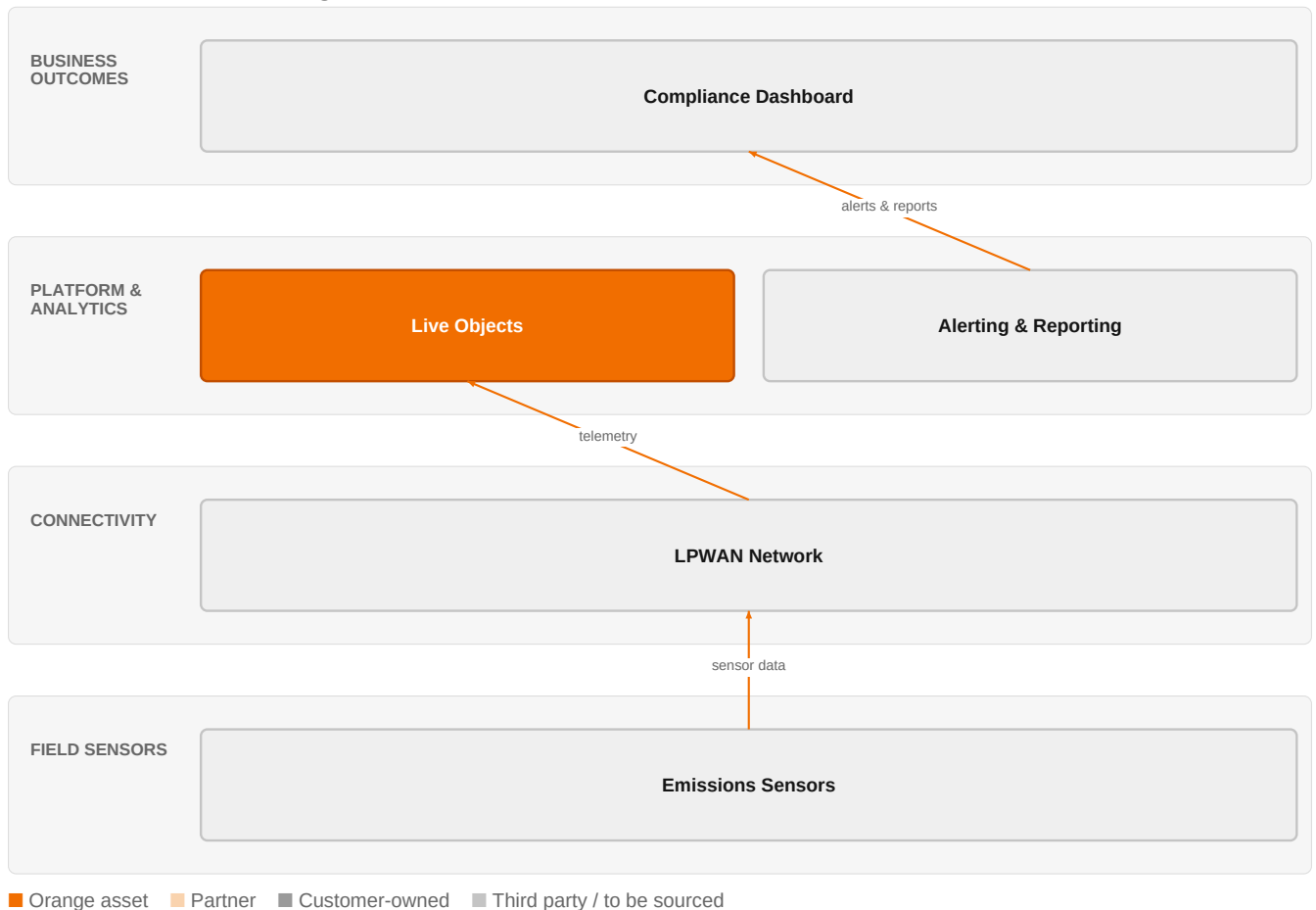
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€220m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 34 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (TH, NG); the estimate covers the European footprint only.

The solution, in one picture

LPWAN Emissions Monitoring Service



This diagram shows how sensor data flows from field devices through LPWAN connectivity into Orange's Live Objects platform, enabling real-time compliance monitoring.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Regulatory pressure is intensifying: Thailand now mandates factory emissions monitoring, and Nigeria's environmental agency is actively assessing gaseous emissions at mining sites. European public authorities are procuring environmental monitoring services, including pollution tracking and marine species monitoring, indicating a broader shift toward proactive environmental compliance. LPWAN technologies (LoRaWAN, NB-IoT, LTE-M) have matured, making low-power, wide-area sensor networks cost-effective for continuous monitoring.

Sources: SIG-234C749A6F8C Envirotech Online 2026-06-09; SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16; SIG-AC86A5D91D50 ted.europa.eu 2026-06-11; SIG-AA6F24D77461 ted.europa.eu 2026-05-27

Who buys, and why now

The buyer is typically a director of environmental compliance or a chief digital officer within a public agency or municipality. The pain is operational: manual emissions inspections are infrequent, costly, and reactive, leaving agencies unable to detect violations in real time or demonstrate compliance to regulators and citizens. The trigger is often a new regulation, a high-profile pollution incident, or a mandate to report emissions data digitally.

Why it is hot now — every claim bound to a dated source

- Thailand tightens air quality rules with mandatory factory emissions monitoring. [SIG-234C749A6F8C]
- OGEPA assesses gaseous emissions and pollutants at a mining site in Nigeria. [SIG-9E90687CF1FE]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a managed service using Live Objects as the IoT platform to ingest and manage sensor data, supported by IoT experts and Digital experts for deployment and integration. The service would include sensor deployment planning, connectivity via LPWAN, data visualization dashboards, and alerting, all hosted on Orange's certified infrastructure (ISO 27001, SecNumCloud, CISPE) to meet public-sector sovereignty requirements. The engagement would start with a pilot in a defined zone, then scale to full coverage, leveraging the Grand Paris Sud reference as a proof point.

Why Orange can win

Orange can win because it combines a proven IoT platform (Live Objects) with sovereign cloud certifications (SecNumCloud, CISPE) that are critical for public-sector data. The Grand Paris Sud reference demonstrates successful deployment in a French public territory. However, the asset list is thin on specific environmental monitoring expertise, so Orange would need to partner for sensor hardware and domain-specific analytics.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
CISPE	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Grand Paris Sud	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.9; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators Telia, Vodafone, and Deutsche Telekom/T-Systems, which offer LPWAN connectivity and compete on network coverage. Actility is a specialist in LPWAN infrastructure. Global systems integrators Accenture, Atos/Eviden, Capgemini, and IBM bring strong public-sector relationships and integration capabilities. The customer will compare Orange's sovereign cloud and platform maturity against these players' connectivity scale or SI expertise.

Sources: SIG-234C749A6F8C Envirotech Online 2026-06-09; SIG-9E90687CF1FE The Guardian Nigeria N 2026-08-16

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Public / Government sector; competes in OX: Smart Industries	structural	—
Atos / Eviden	Global systems integrator	active in Public / Government sector	structural	—

Capgemini	Global systems integrator	active in Public / Government sector; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Public / Government sector; competes in Connectivity Solutions, OX: Smart Industries	structural	—
IBM	Global systems integrator	active in Public / Government sector	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current process for monitoring emissions from regulated facilities, and how often do you conduct inspections?
- 2 Have you received any new regulatory mandates in the past 12 months that require continuous or more frequent emissions data?
- 3 What is the geographic scope of your monitoring responsibility, and are there specific industrial zones or ports where you lack real-time visibility?
- 4 Do you have existing IoT or sensor deployments, and if so, which connectivity technologies are you using?
- 5 What are your data sovereignty and security requirements for environmental data, and do you require hosting within a specific jurisdiction?
- 6 Who in your organization would own the budget for a continuous monitoring service, and what is the procurement timeline?

Objections you will meet

They will say	You can answer
We already have manual inspection processes and they work fine.	Manual inspections are reactive and infrequent, leaving gaps between checks. Continuous monitoring provides real-time alerts and a compliance record that manual methods cannot match, and it reduces the cost of field visits.
We are concerned about the cost of deploying sensors across a wide area.	LPWAN sensors are low-cost and low-power, and the managed service model spreads capital costs into operational expenditure. A pilot can demonstrate the value before scaling.
We are worried about data sovereignty and security.	Orange's infrastructure is certified to ISO 27001, SecNumCloud, and CISPE, ensuring that your data remains within sovereign boundaries and meets public-sector security requirements.

Risks and unknowns

The main risk is that public procurement cycles are long and budget-constrained, potentially stalling sales. There is also uncertainty about sensor hardware availability and calibration standards across jurisdictions. Orange's lack of in-house environmental sensor expertise is a genuine gap that must be addressed through partnerships.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Bid brief: for a public-sector environmental compliance bid, differentiate by combining Live Objects (L1) with ISO 27001 and SecNumCloud certifications, and reference the Grand Paris Sud (SUP) project as proof of successful public-sector IoT deployment.
Sales	In a meeting with a public-sector environmental agency in Thailand, say: 'Orange's Live Objects platform can help you meet mandatory factory emissions monitoring requirements with secure, scalable LPWAN connectivity, backed by our ISO 27001 certification.'
Strategist / Innovator	Deep-dive brief: evaluate how Orange's Live Objects (L1) can support LPWAN-based emissions monitoring for public-sector compliance, leveraging IoT experts (SUP) and Orange Group researchers (SUP) to assess technical fit and regulatory alignment with Thailand's mandatory factory emissions monitoring and OGEPA's mining site assessments.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-AC86A5D91D50	2026-06-11	ted.europa.eu	1	France – Pollution tracking and monitoring and rehabilitation – SPCH a Harbonnieres (80) - Etud...
SIG-AA6F24D77461	2026-05-27	ted.europa.eu	1	Norway – Environmental monitoring other than for construction – Monitoring foreign marine speci...
SIG-9E90687CF1FE	2026-08-16	The Guardian Nigeria News	2	OGEPA assesses gaseous emissions, pollutants at Ogun East Mining Site - The Guardian Nigeria Ne...
SIG-4D1BCBC2C864	2026-07-14	Envirotech Online	2	Mercury emissions monitoring for environmental compliance - Envirotech Online
SIG-234C749A6F8C	2026-06-09	Envirotech Online	2	Thailand tightens air quality rules with mandatory factory emissions monitoring - Envirotech On...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (2 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:17+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:12+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.